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Membrane: a poetics of the border

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ABSTRACT
In this paper, I use the figure of the membrane as a mediating presence between inside and outside that helps to regulate the self against environmental toxicity. Drawing on the biological theory of Lynn Sargulis and Dorian Sagan, Didier Anzieu's theory of the Skin Ego and Ted Chiang's short story 'The Story of Your Life', I trace the appearance of the membrane as a form of material ecocriticism across texts, genres and bodies.

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In Book XI of Ovid's *Metamorphoses*, he tells the familiar story of King Midas, who receives the gift of turning objects into gold from the god Bacchus (Dionysius in the Greek) as a reward for returning Bacchus' companion Silenus to him. 'Make it so that whatever I touch with my body, turns to yellow gold'[1] says Midas, and Bacchus gives him the 'harmful gift'. Midas is initially ecstatic at this new power of his, 'scarcely believing it when he broke off a green twig from the low foliage of the holm-oak: the twig was turned to gold', then picking up stones, clods of earth, apples, corn husks, door pillars and even water, turning them all into gold.

It is only when Midas sits down to a meal prepared by his servants that he realises the foolishness of his gift. As Ovid tells it:

> If he touched the gift of Ceres [the goddess of corn] with his hand, her gift hardened. If he tried, with eager bites, to tear the food, the food was covered with a yellow surface where his teeth touched. He mixed pure water with wine, the other gift of his benefactor, but molten gold could be seen trickling through his lips. Dismayed by this strange misfortune, rich and unhappy, he tries to flee his riches, and hates what he wished for a moment ago. No abundance can relieve his famine: his throat is parched with burning thirst, and, justly, he is tortured by the hateful gold.

Midas then calls out to Bacchus, asking for forgiveness for his sin, and then is told by the god to go to the source of the river Sardis, where he must plunge

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his 'head and body' into the water. There, 'the golden virtue coloured the waters, and passed from his human body into the stream' and Midas was cured.

There are a number of salient points to this story worth noting. At stake is the question of the borders between the self and the outside. In his golden gift, the boundary between Midas' inside and the non-human world of plants, objects and environments has become fixed and static. Midas is unable to manage the exchange between the body and the ecosystems he moves through, and in this construction of a hard border between the self and the others outside, he is unable to survive. Instead of the usual traffic between the human and its outside, which Jane Bennett has so wonderfully described in her explanation of the role of stomach bacteria in eating in *Vibrant Matter*[2], the self makes over the world around it, re-constructing the non-human world of plants and objects in its own image. This is a 'harmful gift', which comes at a profound cost to that self, for 'no abundance' of wealth – fixity – can relieve the need for sustenance and exchange from beyond the border of the body. This is not simply a fable about the flaws of the economy, as many think, for here the economy, in the form of gold, gets *stuck*; it no longer is a flow of exchanges between inside and outside. Gold moves from being an object of use for economic exchange to one used for eating, constructing a body whose borders have become suddenly impregnable, and it is in that movement that it becomes toxic to the self. Land itself goes from sustaining to toxic.

In other words, what is at issue in this story is the question of a *membrane*, the mediating layer that borders the self, creating its form through limitation, as well as moderating the exchanges between that self and the world outside of it. In this essay, I will investigate the membrane as a metonym for dealing with all kinds of otherness, as something that might offer insight into not just how we can distinguish between harm and health, but also into the ways that borders can and do become rigid in ways that prevent the flourishing of the self. Although in the language of reactionary politics, the border is often one that is imagined to produce harm in its openness, critical theorists from Derrida onward have seen the fixity of borders as just as great a harm, a reneging of our fundamental duty towards hospitality.[3] In his 'harmful gift', Midas shows the dangers of too rigid a border, a harm that produces a 'parched throat' from its ability to drink water, but we might note that his immersion in the river Sardis brings with it dangers, too – for we know that humans can die from taking in too much water to themselves just as easily as none at all. In an epoch marked by the proliferation of environmental toxins, we do not have to accept the politics of what Wendy Brown has called the 'rising Right'[4] of new authoritarianisms to recognise that there are some circumstances in which the survival of the self might depend upon closed borders. At the same time, the post-neoliberal Right

in its libertarian approach to the marketisation of social relations demonstrates that not every free exchange is one that sustains.[5]

In other words, it is not simply that either openness or closedness is in itself economically, politically or ethically a social good. What I will argue, therefore, is that the regulation of exchange is something that is at stake in contemporary discussions of the border, and we might learn lessons about how to negotiate the question of the boundary from not only Ovid's Midas but cultural theories and aesthetics of the border from new materialist, psychoanalytic and speculative fiction archives, informed by ongoing dialogue with indigenous thinkers. As the Yolŋu scholars in the Bawaka Collective have shown, our relationship to the environment in which we live can only ever be multi-scalar, multi-temporal, and involve co-becoming with an agential Country.[6]Informed by this knowledge, I will suggest that while the membrane operates differently to the unconditional openness towards otherness at work in deconstructive hospitality[7], in its workings in and across bodies, it vacillates between regulation and exchange, a *semi*-permeable relation to its outside. So while membranes can sometimes provoke disgust reactions in their abjected liminal in-between states[8], I argue they constitute a resource for thinking through the exchange of otherness at work in corporeality that is simultaneously material and semiotic. The membrane is a *figure* in the sense described by Donna Haraway, the point 'where the biological and literary or artistic comes together with all the force of lived reality'.[9]

The membrane

It's one of the founding tenets of new materialism that we are open to the world, constructed, in the words of Stacy Alaimo, trans-corporeally, the site of 'transits and interchanges',[10] porous to the outside,[11] 'congregation entities'[12] cut through with non-human relations. Yet not every intra-action[13] through which we are formed is equally conducive to survival in the age of what Heather Davis has called 'generalized toxicity'.[14] While some interactions and intra-actions are profoundly necessary for continued life, others might harm or even kill organisms or species. To live in the Anthropocene, the era in which humanity has even made an impact on the geological record[15], is to be in a nature-culture[16] that can and harms human and non-human beings. To put it bluntly: some exchanges kill – The Lancet estimates 4.5 million people die annually from air pollution alone[17], while scientists have been tracing the effect of pollution on ecosystems since at least Rachel Carson's work on pollutants in the early 1960s in *Silent Spring.*[18] As Michel Serres[19] has argued, pollution can be a way of marking a spot, appropriating a place for one's own use, and it is bound up in colonial histories and economies.[20] There are times when the

membranes of the human body do not work well enough to filter out harmful pollution or infection, and we turn to such prostheses as the mask.

At the same time, though, emergent[21] fascisms[22] constitute whole communities as forms of toxic pollution, an eliminationism that can just as easily apply to non-human species[23], producing a politics of purity designed to scrub otherness from the social and environmental field.[24] The idea of species as 'weeds' or 'pests' makes possible their removal from ecosystems, just as the designation of the stranger organises eliminationisms around race, gender, sexuality, religion, disability and other signs of visible difference.[25] In purity politics, the membrane is a thick boundary, letting next to nothing through in constructing the self and the state has sealed from the outside. This we might call a *hard border*, one designed to stop the traffic between inside and out, to maintain the ipseity of the organism and seek to eliminate 'foreign' elements. The hard border is one of homogeneity, in which the order of the same rules, and only recognises like. Hard borders have proliferated literally in the form of walls springing up in the United States, Israel and elsewhere[26], and metaphorically, in the nativist right-wing and even centrist politics of those countries and others like Australia. To designate someone or something as pollution is to devalue them, to set in motion the means by which they might be hygienically removed from the world. As we know from the history of genocide, this can be a profoundly dangerous form of thinking, which might legitimate great acts of evil through the banal functioning of a governmentality in which people have become a problem to solve.[27] Moreover, as Ghassan Hage has pointed out, 'racial and ecological domination'[28] are co-extensive with one another, securing the borders and workings of the extractive economies of settler nation-states.

As a result, we can safely say that not every material relation is conducive to life, and not every economy of (non)exchange extends or enriches the flourishing of the self. Here I offer the figure of the *membrane* as a form of 'material ecocriticism',[29] a material text that can be read across biological and social contexts with an irreducible relationship to writing. The OED defines the word membrane as 'any thin, often pliable, sheet or layer, esp. one forming a barrier or lining'. It came into use in the fifteenth century to describe the biological, as 'a thin sheet of tissue or layer of cells, usually serving to cover or line an organ or part, or to separate or connect parts', and soon began in the seventeenth century to be used to describe plants ('A (usually thin) layer of tissue forming part of a plant, seed, etc., esp. as a covering or lining') and human fertility ('The membranes that develop from the ovum to enclose and support a developing vertebrate embryo or fetus; spec. the amnion and chorion in humans'). Etymologically, it comes from the Latin *membrāna*, meaning a membrane (in animal bodies) or parchment.

So there is something in the word that suggests the membrane to be thin, to separate between inside and outside, perhaps between an organism or plant and their environment. As Serenella Iovino has said, bodies are what they are via their permeable boundaries (membranes that cause the flows of energy and matter)' (102). Frog skin, for instance, is a membrane layer of only a cell or two, thus potentially making frogs more vulnerable to the effects of industrial pollutants.[30] In its thinness, a membrane might let through *some* material, but not at all; it works to filter. It's not firm or rigid; it's pliable, but it is not simply agape, for some substances and relations are not allowed through. What is at stake at the border of the membrane is continued survival, not simply of the organism itself but its reproductive capabilities, the ability to protect future organisms by maintaining a boundary that is neither completely open nor completely closed. Derrida once talked about the hymen, whose liminal presence 'sows confusion between opposites and stands between the opposites at once'[31], yet here I want to move beyond that virginal figure to something less gendered, more universal, less strictly anthropocentric. Though at times it operates in an either/or mode, the membrane is often a both/and figure, a gate that can be open and closed at the same time. Here I offer the figure of the membrane as a mediation *in* matter, as a biological poetics articulated through texts both embodied and linguistic.

In the origins of the membrane in parchment, we are thus thrown into a history of writing, in which words as much as substances might be a *pharmakon*[32], a poison or a cure. We might write to construct a boundary (such as that of the nation-state) or to open one up (in the politics of hospitality found in pro-refugee politics), or, in the membrane, meet a transitional point between the two. To write on parchment is to remember the materiality of language, which is now in the computing age as much a 'geology of media'[33] as a biology and chemistry of ink and paper. Ink is minimally toxic, except when ingested, when taken inside the body through swallowing or tattoos – the skin as a membrane for writing. But the contemporary geologics of writing, the computers and phones made from rare minerals and the cloud servers that use up water and fossil fuels, mark out the contemporary terrain of writing as one inextricably bound up in the generalised toxicity of the Anthropocene. As a result, even the status of writing and its effects on ecosystems are manifold and might need to be revisited through the figure of the membrane as a form which mediates the exchange of communication of all kinds between inside and outside. At the same time, we must be alert to the fact that to privilege written text over other forms of conceptualisation of material relations like Australian Aboriginal 'sand talk'[34] is itself the legacy of European colonialism of indigenous peoples, and as Tyson Yunkaporta has argued, to put ideas into writing might in itself warrant forms of protection.[35]

The membrane, therefore, might mark the point through which we filter out the toxic, and though writing has been held out as a purely human endeavour, in bio-semiotic[36] and autopoetic self-narrativising[37], we can trace the construction of membranous boundaries across all kinds of species, human and non-human. Permeability is, in the words of Barad, 'material-discursive'[38], a flow of information and substances across and through bodies and environments, Country, of all kinds. To allow the traffic of some connections and communications, we might need to cease others, without simply building hard borders. It is the case, as Iovino has argued, that 'porosity is not only the basis of change, growth, and decay both on the geological and a human level, but the very condition of history'[39], but so too is the filtering of the harmful from the life-sustaining, and it is these that we must think together as a singled phenomenon co-occurring at any given moment. These are neither ontologically distinct nor entirely separate, relying upon fusion and incorporation as well as rejection and distinction, all at once. Without discarding the critical insights of the new materialisms nor slipping into eliminationist politics, how might we then adjudicate between life-giving and life-harming, between toxic and not?

Membrane-bound

In thinking about the membrane, I want to turn towards the work of the biologist Lynn Margulis, who has theorised that evolution emerges largely through symbiosis, through what she terms ''symbiogenesis'.[40] While evaluating the scientific validity of Margulis' theories of evolution and the Gaia theory is beyond the scope of scholarship in the humanities, I follow the likes of Donna Haraway[41], Isabelle Stengers[42], and Bruno Latour[43] in seeing in her work useful conceptual resources for understanding biological processes from the cellular to the planetary that might allow us to recalibrate our relationship to material life. In these, the figure of the membrane features heavily as a border that contains, produces and mediates exchange between inside and outside.

On their *Origins of Sex,* Margulis and her son Dorion Sagan have argued that eukaryotic cells, which are complex, organised cells bound by a membrane identified by the presence of a nucleus and various organelles, are 'homologous to a community of microorganisms'.[44] Margulis and Sagan see the eukaryotic cell as the sum of its history of bacterial precursors distributed across the cell rather than strictly found in its nucleus (as some theories of inheritance have it). Distinct from a Mendelian reproduction of species marked by the genetic inheritance of organisms from their progenitors, Margulis and Sagan see a more chaotic role for the evolution of species marked by the co-existence of otherness and the exchange, interaction, fusion and repulsion of difference across borders. This is a distinctly queer ecology in

the sense that Timothy Morton has described, one in which 'all life-forms, along with the environments they compose and inhabit, defy boundaries between inside and outside at every level'.[45] Yet though Morton describes a defiance of heteronormative ideas of capital n 'Nature' through a queer traffic between inside and outside, this does not mean that there are never any borders, nor bounded (in some sense, at least) compositions.

Margulis and Sagan, in fact, argue for the key role of the membrane to give form to the eukaryotic cell (as distinct from a prokaryotic cell, which lacks a nucleus and organelles). Eukaryotic cells are found across different forms of life – protozoa, fungi, plants, and animals – and have internal plasma membranes that divide the organelles from one another inside the cell, as well as a membrane to the outside. Margulis and Sagan argue that:

> All eukaryotic individuals must reserve, in a form capable of continued reproduction, their genetic components, the remnant bacteria in the combined form of the nucleocytoplasmic, mitochondrial, plastid, and undulipodial genomes. If we accept the cell as a microbial community, the germ plasm is equivalent to component autopoiesis: a complete set of heterologous genomes and their protein synthetic systems contained within a membranous package – not the nuclear membrane but the plasma membrane.[46]

The eukaryotic cell is thus marked by its internal difference, of a history of otherness and exchanges inside and out that have been taken across multiple membranes, including their border with the outside. This membrane gives it form, and it is vital to this theory that there *be* a membrane – the word membrane is usually 'membrane-bound' in Margulis and Sagan – to divide a cell from its outside. To simply *be*, an organism needs what Haraway terms 'mortal boundary formations',[47] borders entangled with the vulnerability of the organism.

As Margulis and Sagan note, though, it is just as vital that there be exchange across membranes for evolution to take place. A membrane-bound organism is not the same as the bacteria spirochetes, though Margulis postulates that spirochetes' ancestors 'invaded' archaebacteria to become centriole-kenetosome eukaryotic nucleated cells.[48] This invasion is one of the instances in which fusion did occur, but this is not universal in Margulis and Sagan's work, despite the high priority they place on the chaotic integration of bacteria in evolving life. 'Attraction, merger, fusion, incorporation, cohabitation, recombination – both permanent and cyclical – and other forms of forbidden couplings, are the main source of Darwin's missing variation'[49] in his theory of evolution. These processes are not precisely synonyms – attraction and cohabitation imply that becoming-with does not in every circumstance mean simply the fusion of all with all, but rather, the legacy of singular histories of meeting and co-evolution of disparate elements, of distinct crossings. In those crossings, which do not constitute

the whole of reproduction but rather its leading edge of encounters with otherness, as Haraway notes, there are quite a lot of 'acidic reminders of mortality',[50] of failure and precarity.

Though it does not appear in the *Origins of Sex*, published in 1986, over time, Margulis, both separately and with Sagan, began to link this theory with her other great innovation, the Gaia theory developed with James Lovelock. This occurs most notably in 1998s *Symbiotic Planet*, which moves from the cellular to the planetary. She states that 'the surface of the planet, Gaia theory posits, behaves as a physiological system in certain limited ways',[51] including the regulation of temperatures, acid-alkalinity and reactive gases, as well as co-evolving in and across species. Though the theory often has vaguely New Age connotations[52], in part a legacy of its etymological links to the earth goddess Gaia, Margulis intended the theory to be scientifically rigorous and plausible. Bruce Clarke summarises the implications of her movement from the cellular to the planetary for ecological thinking:

> Ecological communities themselves have shifting but relatively stable identities. Site- specific, they endure and mature. However, even while their living components carry on their reproductive ways, insofar as they also maintain their individual autopoieses and contribute them to a group dynamic, such communities do not reproduce their organization in the form of progeny. Rather, their continuity over time must emerge from the composite maintenance of the ecosystemic consortium.[53]

In other words, Clarke describes how autopoesis requires both a certain fixity and a co-evolution with other organisms rather than through reproductive progeny. The intra-action with the outside at work in symbiogenesis folds into the self-composing of the cell, and scales up from there. Margulis' version of the Gaia theory thus moves us from the cellular up to the planetary, as a set of principles of exchange that moderate and mediate the continued evolution of life on this planet.

I argue, therefore, that Margulis and Sagan's theories of co-evolution show us how the emergence of complex life depends upon exchange and the taking-in from the outside of certain kinds of difference, in the form of bacteria. At the same time, though, the cellular membrane works to give form to the eukaryotic cell, creating a border that does not let everything. Margulis states that 'in certain cases cohabitation, long-term living, results in symbiogenesis'[54] – not all. The cellular membrane is, in the terms of Haraway, a 'contact zone'[55] that borders and mediates and only lets through *certain* symbiotic relations – a selectivity that often goes unspoken in Margulis. To cohabit, to live with, might result in co-evolution, but it might not. Margulis talks about how 'genes are a palpable legacy of a violent, competitive, and truce-forming past'[56], a martial metaphor that takes for granted the disappearances of a myriad of forms of otherness even as it

makes space for co-existence among difference. As Haraway has put it, 'to be one is always to *become with* many'[57], and in the workings of the cell as much as the planet, we find the sense that we are both singular and plural at the same time.

Thinking through the skin

In thinking about the membrane, we inevitably come into contact with the skin, with those surfaces of bodies that bear the traces of contact from other bodies and non-human agencies. Writing in the late 1980s, the French psychoanalyst Didier Anzieu[58] offered the idea of the 'skin-ego' as a metaphor for the self, setting this in an environmental frame in which patients can no longer 'perceive the frontiers between the psychical and bodily Egos'[59] – that is to say, recognise the boundaries between inside and outside with any certainty. Anzieu situates his research within a context whereby social relations created through communicative technologies and 'violence against nature'[60] in the form of widespread pollution contribute to the 'suffering' of his patients. Anzieu formulates his task in the following way:

> to rebuild limits, restore frontiers and create for ourselves recognisable and habitable territories. These limits and frontiers must both institute differences and enable exchanges between the newly defined regions – regions of the psyche, of knowledge, of society and humanity.[61]

In setting up the skin as an interface between inside and outside through which the self is produced, Anzieu sees his work as analogous to the move among biologists towards examining the outer membrane of cells[62], as well as post-Atomic age physicists. To be permeable is to be always in some way vulnerable, able to be affected in negative as well as positive ways.

Anzieu's wager is that Freud's theory of the ego, the self, far from being marked by a psychic depth, can be found at the surface, at the skin. He says, 'what if thought were as much a matter of the skin as of the brain? And what if the Ego, now defined as the Skin-Ego, had the structure of a wrapping?'.[63] In what the trans theorist Jay Prosser terms the 'wonderfully uncomplicated literalism' of Anzieu's work, 'the body is crucially and materially formative of the self'.[64] For Prosser, this implies that the corporeal changes of transition fundamentally change a transperson's psyche in a dialectical movement between body and mind, and here I want to extend this work by suggesting that we might see the skin-Ego as a membrane whose refiguration of embodiment can be repurposed to respond to environmental and social differences – including the toxicity that pollution often entails – of all kinds. Anzieu does not suggest that inside and outside are the same, nor that our entanglement with difference entails the becoming-same through

contact. Rather, in what he calls the 'consensuality' (consensualité) of the Skin-Ego[65], he installs the production of the self at the meeting point of the membrane, whose limits and frontiers are necessarily *semi* permeable – letting through exchanges while also setting boundaries.

For Anzieu, the skin has three main roles for the self, which are grounded in biological functioning:

> The first function of the skin is to be the sac that contains and retains inside itself all the good, full material that has accumulated through breastfeeding, everyday care, and the experience of being bathed in words. Its second function is to be the interface that marks the border with the external world, which it keeps on the outside, the barrier that protects one against being penetrated by the aggression and greed of others, whether people or objects. The third function of the skin, which it shares with the mouth and carries out at least as much as the mouth does, is to be a site and primary mode of communication with other people, to establish meaningful relations; in addition, it is a surface for registering the traces left by those others.[66]

Each of these touches on the role of the membrane – to give form by constructing a layer, to mediate relations between inside and outside, and to communicate with that outside, as well as the role of language and desire in that mediation. We must have some sense of form to be, and the membrane ensures that there are boundaries to the body. At the same time, exchanges with the human, and indeed the non-human, construct the boundary of the skin and, in turn, formulate the resulting sense of self. Far from being solely a natural phenomenon, the skin emerges as a natural-cultural artefact, for which the membrane regulates the traffic of substances and language between the self and its environment. The Skin-Ego is thus profoundly ecological, a material-discursive structure for which the agency of the non-human is necessarily entangled with the makings of the self.

The Skin-Ego is not a hard border; it lets through some but not all other beings and substances, though Anzieu suggests that the role of the membrane is to adjudicate against aggression and greed (and here we are reminded of Midas' legendary greed). In describing this *as* greed, Anzieu may use an overly moralising tone, but here I want to suggest there might be an anti or at least non-capitalist membrane, which adjudicates between exploitation and extraction or violence from human and non-human alike. In giving such a role to 'objects', Anzieu is positing a kind of inhuman violence which may or may not be wielded by a person, which may or may not be motivated by emotions, but it might also be situated within the workings of broader, more impersonal, material, economic, affective and political structures. Here Anzieu's work can be compared to the actor-network theory of the likes of Bruno Latour[67] as a darker iteration that makes space for negative or dangerous forms of otherness against which the self might have to protect itself. In other words, there is a sense that the

membrane might function as a protective layer against not merely individual but structural otherness. At the same time, though, the Skin-Ego remains open to the outside in some sense, a surface that is marked by its relations to others.

Here I think we can take Anzieu further in thinking through the implications of the Skin-Ego, for whom the 'wrapper' or 'container' of skin does so much work. There is at times a somewhat conservative gender and sexual politics to Anzieu, who writes of how 'the vagina is not an organ with a specific structure but a fold of the skin [. . .] whose mucous membrane is exposed, so that its sensitivity and erogeneity are on the surface', yet he seems to unnecessarily restrict this to a cis-normative and heteronormative paradigm when he describes the vagina 'reaching a climax on contact with the equally sensitive surface of the tip of the erect penis'.[68] Here, we might recall the associative link between the membrane and reproduction, the ways that the boundaries of bodies necessarily involve sexuality and fertility. Yet as Prosser has shown, Anzieu's work offers a rich array of possibilities that move us beyond the cis and heteronormative, as well as opening us to experiential knowledge of land. If the self emerges as a result of the contact across membranes, there is no reason to assume that this necessarily involves cisgender or heterosexual bodies or contact, or that the kinds of communication that are mediated by membranes are necessarily only human.[69] Though the membrane as a figure involves eroticised communication, we cannot and should not assume the teleological use of bodies towards what Lee Edelman once called heterosexual 'reproductive futurism'[70], nor should we necessarily place those uses within anthropocentric frames that do not account for the agency of the non-human. The self of the Skin-Ego can be constructed in a myriad of approaches of difference across the membrane to human and non-human, or as Yunkaporta has put it, 'the real knowledge will keep moving in lands and Peoples, and I'll move on with it'.[71]

In thinking about the Skin-Ego through this broader frame, which is inspired by Prosser's formulation of transitioned bodies as constructing trans selves, I am trying to bring Anzieu into a more-than-human frame informed by indigenous thinking, to recalibrate what it is that we imagine that bodies and psyches might do and be to think through our relations to generalised toxicity. In positing the 'wrapping' of the semi-permeable skin, Anzieu allows us to understand that our relationship to the material world is never entirely outside, that it is always folded within the body as well. To touch is to fold, yet this is not simply fusion, even as our skin is marked by its histories of approach and contact[72], of aggression and care, and language and touch. Our bodies thus can and do bring inside themselves the life-extending and the life-diminishing, the fertile and the toxic – all of which are mediated by the membrane. To investigate this figure of the

semi-permeable membrane further, I now turn to the science fiction novella *Story of Your Life* by Ted Chiang.

Story of your life

For Anzieu, a work of art can be considered something of a membrane in itself, 'a skin that holds together sensuality, motivity [motricité] and affectivity, a wrapping that unifies momentarily the past, the present and the expectation of a future, a membrane that harmonises the vibrations of the body with the internal rhythm of the code'.[73] Ted Chiang's 2000 novella *Story of Your Life*[74] is such a text, a narrative about first contact with an alien species called heptapods, with whom a linguist named Louise is contracted by the Army to facilitate meaningful communication through learning their language, 'a nonlinear system of orthography that qualified as true writing'.[75] What is interesting about this encounter is that it occurs through a membrane, nicknamed a 'looking glass' in the text.[76] These act 'as two-way communication devices, presumably with the ships in orbit. No one knew why the aliens wouldn't talk to us in person; fear of cooties, maybe'.[77] Chiang raises the possibility of the viral, of an exchange that might harm as likely as prompt the evolution discussed by Margulis and Sagan.

The looking glass is a transitional object, mediating between the human and the alien. This is described thus:

> As we crossed the paint line, the looking glass appeared to grow transparent; it was as if someone was slowly raising the illumination behind tinted glass. The illusion of depth was uncanny; I felt I could walk right into it. Once the looking glass was fully lit it resembled a life-sized diorama of a semicircular room. The room contained a few large objects that might have been furniture, but no aliens. There was a door in the curved rear wall.[78]

A looking glass allows us to see, but it also protects, it mediates. As a membrane, it borders between two different states, but it's thin enough to let light through, so that the outside can be seen through the inside. It might protect the body from the aggression or greed of others, but it might also work as a site for contact, enabling exchanges with others in ways that are always materially embodied, economic, and political.

It becomes clear that the heptapods need the looking glass membrane to communicate because the conditions they require for life are fundamentally different to Earth. To be able to breathe, as Jean-Thomas Tremblay[79] has shown, depends upon a fundamentally conducive set of social as well as environmental conditions. Louise describes it thus:

> The heptapods' planet had two moons, one significantly larger than the other; the three primary constituents of the planet's atmosphere were nitrogen,

> argon, and oxygen; and fifteen twenty-eighths of the planet's surface was covered by water. The first words of the spoken utterance translated literally as 'inequality-of-size rocky orbiter rocky-orbiters related-as-primary – to-secondary.[80]

The narrative estranges the land as dwelling, becoming a relation as much as a place. Here, we need the membrane to communicate across differences, to make equivalents across the unbridgeable. Co-existence means not simply the changing or fusion of the self, but rather, the mediation of the ways that language works across the membrane to communicate between incommensurable differences and relations to land. For humans, the skin shares with the mouth the ability to speak, but in this story, the membrane takes precedence in the contact between different selves.

Time, too, is a kind of membrane in this narrative, mediating between a diffractive past, present and future – Louise describes how 'for the heptapods, all language was performative. Instead of using language to inform, they used language to actualise'[81]. Action thus confirms at the same time as it creates through the performative, an always-already that is also an always-will-be. The aliens live in an everywhen[82], where 'sure, heptapods already knew what would be said in any conversation; but for their knowledge to be true, the conversation would have to take place'.[83] Rather than the meeting between externalities of interaction, we are folded into the world through intra-action, in which we are already inside our own actions. Sometimes the membrane lets through, and here the heptapods, who have come 'to see', or 'to observe' through the looking glass, have mediated what has occurred, a material-discursive performative in which the agency of the non-human matters.

The participatory observation of the heptapods is framed by Chiang through a notion of causality referencing the seventeenth-century Fermat's Law, which links wave and ray optics. In this everywhen, the future is known equally as the past to the heptapods, but it must nevertheless be enacted to be. The narrative reproduces this law, being addressed to Louise's adult daughter but ultimately ending up with her conception – an action that must take place for the narrative in the past to occur, even as it brings to mind the painful ultimate separation between Louise and her husband, Gary. Union and separation occur equally here in an entangled intra-action across shifting temporal scales – this is no paean to unmediated unity with the universe – the membrane function remains throughout past, present and future. In other words, as with Anzieu's Skin-Ego, the self is produced at the meeting point of the membrane with the outside world, with its histories of contact with an otherness that is always mediated by and through that membrane.

In the end, the looking glass turns out to be surprisingly mundane in its composition:

> That final 'gift exchange' was the last we ever saw of the heptapods. All at once, all over the world, their looking glasses became transparent and their ships left orbit. Subsequent analysis of the looking glasses revealed them to be nothing more than sheets of fused silica, completely inert.[84]

There's nothing otherworldly about the membrane that separates alien from human – rather, it is the relation between the two via the mediating function of language that makes it important. Without the membrane, communication is strictly impossible, but with it, both the traffic of energy and matter can take place, as well as the filtering out of the potentially harmful. Without a living membrane mediating the exchanges between the two beings, two cultures and their entwined social structures, we have an 'inert' materiality without movement. We are returned to a world of objects that appear suddenly lifeless, without the more-than-human linguistic exchange between selves – simply, a hard border of fused silica.

Conclusion

In the story of King Midas, we can see the dangers of the hard border, which might impede our ability to make the kinds of exchanges with the outside needed to survive and thrive. There, the membrane does not work, and the results become unlivable for the King, who must repent his 'sin' of greed and extreme separation. Hard borders can harm and prevent us from engaging in the traffic with otherness that Margulis and Sahan argue drives the evolutionary changes of species. Yet we cannot remain entirely open to the outside either, lest we be harmed when the environment outside becomes polluted and toxic, in which we might not be able to breathe in such an atmosphere, as for the heptapods of Chiang's story. Unmediated openness can compromise the ipseity of the self, so that the self no longer *is* a self. What we can see in both these narratives and in Anzieu's theory of the Skin-Ego is how the membrane functions as a mediating influence. In the exchanges with an outside mediated through the membrane, then, we can find ourselves newly made over by an otherness that has changed who we were from the start.

But where historically, post-Enlightenment cultures have often prioritised transparency and openness as the goal of contact and communication, in the membrane, we might be somewhat opaque to one another, and it is only through that opacity that touch and communication become possible in a pervasively polluted world. As Édouard Glissant has argued, exchanges should be marked by a certain opacity, a letting be of the difference of others and their heterogeneity from the self.[85] We might also expand this opacity to sexual exchange beyond the cis-heteronormativity of Anzieu, as Avgi Saketopoulou suggests, to see the desire that animates sexual traffic across membranes as a both/and proposition, fusing and distinguishing at

the same time as it is propelled by a desire whose workings are always opaque and partially unconscious to us.[86]

The membrane thus works across different material and temporal scales, a relation that is always ecological as well as economic, that mediates the relations between different bodies, cells and structures. In a beautiful passage in *Symbiotic Planet*, Margulis suggests that the planet itself is a kind of body:

> Just as the human body is sharply bounded by skin, temperature differences, blood chemistry, and a calcium phosphate skeleton, so is Earth distinguished from its surroundings by its persistently anomalous atmosphere, its steady temperature and its unusual limestone and granitic rocks [. . .] The planet's surface is not just physical, geological, and chemical, or even just geochemical. Rather, it is geophysiological: it displays the attributes of a living body composed of the aggregate of Earth's incessantly interactive life.[87]

There is no thriving without the membrane, which might be cellular, bodily or even planetary. Without some kind of exchange, evolution might come to a halt; with too much, the individual or species might not survive. To become stuck as King Midas does might lead to death if not cured, if the membranes of exchange are not reopened, though the permeability of the membrane always raises the possibility of the virus in ways that might harm as well as prompt our evolution. In the end, we do not need to see one another fully, nor merge or fuse completely to be in contact – though we might – and it is through the adjudications of the membrane we find that we are always-already bound up in economies of exchange of language, desire, and an embrace with each other and the more-than-human animate land that mediation makes possible in the first place. We are semi-permeable, wonderfully, terrifying, so.

Notes

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4. Wendy Brown, *In the Ruins of Neoliberalism: The Rise of Antidemocratic Politics* (New York Columbia UP, 2019), pp. 2.
5. Adam Kotsko, *Neoliberalism's Demons: On the Political Theology of Late Capital* (Stanford: Stanford UP, 2018).
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8. Julia Kristeva, *Powers of Horror: An Essay on Abjection*, trans. Leon S. Roudiez (New York: Columbia UP, 2024).
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11. Serenella Iovino, "Bodies of Naples: Stories, Matter, and the Landscapes of Porosity," *Material Ecocriticism*, ed. Serenella Iovino and Serpil Oppermann (Indianapolis: Indiana UP, 2014), pp. 97–113.
12. Bennett, pp. 34.
13. Karen Barad, *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning* (Durham: Duke UP, 2007).
14. Heather Davis. *Plastic Matter* (Durham: Duke UP, 2022), pp. 2.
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18. Rachel Carson, *Silent Spring* (New York: Penguin, 1999).
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20. Max Liboiron, *Pollution Is Colonialism* (Durham: Duke UP, 2021).
21. Albert Toscano, *Late Fascism: Race, Capitalism and the Politics of Crisis* (New York: Verso, 2023).
22. Jason Stanley, *How Fascism Works: The Politics of Us and Them* (New York: Random House, 2018).
23. Deborah Bird Rose describes how flying foxes have been constituted as a form of pollution in New South Wales and Queensland in Australia, leading to noise pollution – torture – against populations by local civic authorities, despite their protected status as an endangered species. Deborah Bird Rose, *Shimmer: Flying Fox Exuberance in Worlds of Peril* (Edinburgh: Edinburgh UP, 2022).
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32. Derrida, *Dissemination*.
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36. Jesper Hoffmeyer, *Signs of Meaning in the Universe* (Indianapolis: Indiana UP, 1996).
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38. Barad, *Universe*, 45.
39. Serenella Iovino, "Bodies of Naples: Stories, Matter, and the Landscapes of Porosity," *Material Ecocriticism*, ed. Serenella Iovino and Serpil Oppermann (Indianapolis: Indiana UP, 2014), pp. 102.
40. Lynn Margulis, *Symbiotic Planet: A New Look at Evolution*. Perseus, 1998, 33.
41. Donna Haraway, *Staying with the Trouble: Making Kin in the Chthulucene* (Durham: Duke UP, 2016) and *When Species Meet*.
42. See her riff on the Gaia thesis in *In Catastrophic Times: Resisting the Coming Barbarism* (London: Open Humanities Press, 2015).
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44. Lynn Margulis, and Dorion Sagan, *Origins of Sex: Three Billion Years of Genetic Recombination*. Yale University Press, 1986), pp. 170.
45. Timothy Morton, "Queer Ecology." *PMLA* 125, no. 2 (2010): 274.
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47. Haraway, *When Species Meet*, pp. 31.
48. Margulis, *Symbiotic Planet*, 43.
49. Lynn Margulis, and Dorion Sagan, *Acquiring Genomes: A Theory of the Origins of Species* (New York: Basic Books, 2002), pp. 205.
50. Haraway, *When Species Meet*, pp. 31.
51. Margulis, *Symbiotic Planet*, pp. 123.
52. Bruno Latour and Timothy M. Lenton have argued that the Gaia theory has "either been embraced with too much enthusiasm or rejected with too much skepticism, both without unpacking its exact content" (660) of the self-regulation of planet. Bruno Latour, and Timothy M. Lenton, "Extending the Domain of Freedom, or Why Gaia Is So Hard to Understand." *Critical Inquiry* 45, no. 3 (2019): pp. 659–80.
53. Bruce Clarke, *Gaian Systems: Lynn Margulis, Neocybernetics, and the End of the Anthropocene*. (Minneapolis: University of Minnesota Press, 2020), pp. 165.
54. Margulis, *Symbiotic Planet*, pp. 33.
55. Haraway, *When Species Meet,* pp. 4.
56. Margulis, *Symbiotic Planet*, pp. 37.
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58. Anzieu has been relatively neglected, with the most notable response to his work Naomi Segal's monograph on gender and Anzieu. Naomi Segal, *Consensuality: Didier Anzieu, Gender and the Sense of Touch* (Leiden: Brill, 2009).
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60. Anzieu, *Skin-Ego*, pp. 7.
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64. Jay Prosser, *Second Skins: The Body Narratives of Transsexuality* (New York: Columbia UP, 1998), pp. 65.
65. Segal, *Anzieu*, pp. 5.
66. Anzieu, *Skin-Ego*, pp. 44.
67. Bruno Latour, *Reassembling the Social: An Introduction to Actor-Network-Theory* (Oxford: Oxford UP, 2005).
68. Anzieu, *Skin-Ego*, pp. 10.
69. For an interesting take on the eroticism of the non-human, see Ellen van Neerven's story "Water", in which a queer Australian indigenous person falls in love with a sentient plant being. I'm also thinking of Darcey Steinke's suggestion that post-menopausal cis women might connect to non-human beings like orca whales rather than cis-heterosexuality. Ellen van Neerven, "Water." *This All Come Back Now: An Anthology of First Nations Speculative Fiction*, ed. Mykaela Saunders. (Brisbane: UQP, 2023), pp. 221–46, and Darcey Steinke, *Flash Count Diary: Menopause and the Vindication of Natural Life* (New York: Sarah Crichton, 2019).
70. Lee Edelman, *No Future: Queer Theory and the Death Drive* (Durham: Duke UP, 2004), pp. 5.
71. Yunkaporta, *Sand Talk*, pp. 20.
72. Here I'm also thinking of Sara Ahmed, *Queer Phenomenology: Orientations, Objects, Others* (Durham: Duke UP, 2006).
73. Cited Anzieu, *Skin-Ego*, pp. 29.
74. Ted Chiang, "The Story of Your Life." *Stories of Your Life and Others* (London: Picador, 2015).
75. Chiang, "Story," pp. 13.
76. This in Darko Suvin's (1979) terms constitutes the "novum" of the text, its original twist on consensual reality. Darko Suvin, *Metamorphoses of Science Fiction: On the Poetics and History of a Literary Genre* (New Have: Yale UP, 1979).
77. Chiang, "Story," pp. 4.
78. Chiang, "Story," pp. 5.
79. Jean-Thomas Tremblay, *Breathing Aesthetics* (Durham: Duke UP, 2022).
80. Chiang, "Story," pp. 22.
81. Chiang, "Story," pp. 34.
82. I take the term "everywhen" from Australian indigenous writer Mykaela Saunders. Mykaela Saunders, "Overture," *This All Come Back Now: An Anthology of First Nations Speculative Fictions*, ed. Mykaela Saunders, (Brisbane: UQP, 2022), pp. 1–22.
83. Chiang, "Story," pp. 34.
84. Chiang, "Story," pp. 38.

85. Édouard Glissant, *Poetics of Relation,* trans. Betsy Wing (London: Penguin, 2025).
86. Avgi Saketopoulou, *Sexuality Beyond Consent: Risk, Race, Traumatophilia* (New York: New York University Press, 2023).
87. Margulis, *Symbiotic Planet*, pp. 123.

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